Model selection and parameter description

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1. Model class

To catch the dynamic of each PCI point, I find the best possible order and the best selection of exogenous variables that minimize the prediction error on previously unseen data for the following model:

$$y_t = \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{j=1}^{q} \theta_j \epsilon_{t-j} + \sum_{k=1}^{exo} \beta_k x_{tk} + intercept + \epsilon_t$$

where y is the target variable, ϵ_t is the model error at a specific time step (also called innovation variance or random shocks), x_{tk} is an exogenous variable, and ϕ, θ and β are parameters that are fit during model training (alongside with the intercept). p, q denote the order of the model (how far back do we look into the past, while exo denotes the number of exogenous variables used.

2. Exogenous variables

2.1 Fourier terms

$$x_{k,t} = (sin(2\pi k \times \frac{t}{365.24})), cos(2\pi k \times \frac{t}{365.24})))$$

Used to account for possible yearly seasonality. For every order k, we include both terms in the model. Not necessarily used in the optimal model (i = 0).

2.2 Temperature and Waterlevel

The corresponding daily air temperature and water level mean are included for every PCI point measurement. Orders > 0 describe how many past timesteps should be included.

2.3 Temperature and Waterlevel mean of last n days

Since the resolution of the PCI points is six days, the air temperature and water level mean of the past n days can also be considered. Here only one variable was tested (Either 0, 3, or 6-day mean, not necessarily used in the final model).

3. General model selection pipeline

All possible models were fitted to 90% of the available data and scored on the remaining 10%. The model with the lowest MSE was selected as the best fit. After refitting a model with the selected specifications on 100% of the data, this model was used to forecast the years 2020 and 2021 (depending on the training data reach). For consistency, the Sarimax class of statsmodels was used to handle all possible model cases. In total, 810 models were tested for each available PCI point individually (Temperature and water level variables are sharing their order to keep the search space reasonable). The best model was selected independently for every point (the dynamic of different points can differ quite a bit). Data is normalized during the model fit. Outliers were removed via detrending and boxplot filtering. Predictions are renormalized to the original range. The complete pipeline is available Here.

4. Exact model search space (limits included):

Parameter	Low range	Upper range
AR term	0	4
Differencing term ($y_t^{'} = y_t - y_{t-1}$)	0	2
MA term	0	1
Fourier terms	0	2
Temperature and Waterlevel term	0	2
Mean Temperature and Waterlevel of the n last days	0	6